University of Bolton

MODULE DATABASE

2025-26 > Onwards

# COM7411 (Validated in 2025-26)

## Research Methods

| | |
|---|---|
| **Credits** | 20 |
| **Level** | HE7 |
| **Type** | Standard |
| **Trimester 3?** | Yes |
| **ECTS** | 10 |
| **Marking Scheme** | Numeric mark scheme (i.e. mark of 0-100) |
| **Pass Mark** | 50% |
| **Delivery Type** | This Module is delivered by flexible and distributed learning (FDL) methods, which do not require you to attend particular classes or events at particular times, in particular locations. |
| **Module Outline** | This module develops advanced research knowledge and skills essential for postgraduate study in computing and related disciplines. It enables students to design, conduct, and evaluate ethical and methodologically sound research while developing confidence as independent investigators. The module draws on contemporary research practices and debates to ensure relevance to today's technological and professional contexts. Flipped learning is used to augment engagement, allowing students to explore preparatory materials before scheduled virtual workshops focused on application and discussion. The module complements and augments all others in the programme by strengthening students' ability to apply critical thinking, research design, and evidence-based analysis across diverse contexts. Through an applied portfolio, students integrate theoretical understanding with practical research design, consolidating learning through reflection, ethics, and professional communication. |

### Indicative Content

| | |
|---|---|
| 1 | Foundations of research philosophy and design |
| 2 | Quantitative, qualitative, and mixed-methods approaches |
| 3 | Sampling and data collection strategies |
| 4 | Data analysis and interpretation techniques |
| 5 | Ethics, governance, and professional conduct in research |
| 6 | Academic writing and dissemination of findings |
| 7 | Use of technology in research (e.g., data analysis tools, referencing software, AI-assisted methods) |
| 8 | Reflection, evaluation, and the assessment of research quality |

### Learning Outcomes

On successful completion of this Module you will be expected to be able to:

| | |
|---|---|
| 1 | Critically evaluate a range of research methodologies and philosophical approaches relevant to computing and data-oriented disciplines. |
| 2 | Design and justify an ethical and feasible research proposal informed by critical engagement with relevant literature. |
| 3 | Synthesise research findings through appropriate analysis, interpretation, and academic communication. |

**Learning And Teaching Strategy**

Learning and teaching combine applied research activities, collaborative discussion, and guided study to develop advanced research knowledge and skills. Online lectures, seminars, and case-based discussions introduce key concepts, research methodologies, and academic practices relevant to computing disciplines. Flipped learning enhances engagement by providing preparatory materials in advance, enabling active participation in scheduled sessions. Interactive workshops and structured activities enable students to design, evaluate, and refine research proposals, critically analyse literature, and apply appropriate methodological approaches to practical and ethical challenges. Tutor-led discussions and peer review activities support the development of critical thinking, academic argumentation, and methodological justification, linking theory with research practice. Collaborative activities promote scholarly discussion and shared evaluation, while guided independent study and directed reading support deeper engagement with research literature and the development of methodological rigour. Formative feedback

throughout proposal development strengthens critical analysis, academic communication, and reflective research practice.

### Learning & Teaching Methods

| Method | KIS | Hours |
|---|---|---|
| Scheduled | Scheduled | 45 |
| Independent | Independent | 155 |
| **Total** | | 200 |

**Continuous Assessment Strategy**

Continuous assessment underpins learning throughout the module, ensuring sustained engagement with research design and development. Formative activities, such as proposal outlines, literature critiques, and feedback workshops, enable students to refine their research ideas and methods in response to tutor and peer feedback. These formative tasks contribute directly to the development of the final portfolio. The summative assessment consists of a portfolio of coursework, demonstrating the ability to design, justify, and communicate a research proposal supported by ethical awareness and critical reflection. This structure allows students to evidence academic integrity, professional communication, and readiness for the subsequent project module.

### Summative Assessments

| | Assessment | KIS | Description | | Learning Outcomes | Marking Scheme | Passmark | KIS Weighting |
|---|---|---|---|---|---|---|---|---|
| 001 | Portfolio | Coursework | A portfolio culminating in a report detailing the research background, critical review of relevant literature, and proposed methodology, supported by an online presentation that discusses and defends the proposed investigation within a computing context. | FINAL | 1 2 3 | Standard University Regulations | - | 100% |

### Learning Resources

#### Essential Reading

Cohen, L., Manion, L. and Morrison, K. (2018) Research Methods in Education (8th ed.). Milton Park: Routledge, Taylor & Francis Group.

Creswell, J. W. and Creswell, J. D. (2022) Research Design: Qualitative, Quantitative, and Mixed Methods Approaches (6th ed.). Thousand Oaks: SAGE Publications.

Gray, D. E. (2025) Doing Research in the Real World (6th ed.). London: SAGE Publications.

Hammond, M. and Wellington, J. (2020) Research Methods: The Key Concepts (2nd ed.). Milton Park: Routledge, Taylor & Francis Group.

**Recommended Reading**

Field, A. P. (2022) An Adventure in Statistics: The Reality Enigma (2nd ed.). Thousand Oaks: SAGE Publications.

Hammond, M. and Wellington, J. (2020) Research Methods: The Key Concepts (2nd ed.). Milton Park: Routledge, Taylor & Francis Group.

Mbanaso, U. M., Abrahams, L. and Okafor, K. C. (2023) Research Techniques for Computer Science, Information Systems and Cybersecurity. Berlin: Springer Nature.

Talis Aspire Reading lists

| | |
|---|---|
| **Additional Teaching, Learning and Assessment arrangements for asynchronous delivery of this module only** | Scheduled learning is delivered through synchronous online sessions, including live lectures, seminars, and workshops. Asynchronous learning is used to support and extend engagement, including access to recorded materials, guided readings, discussion forums, and preparatory or consolidation activities that complement the scheduled sessions. |
| **Feedback to Students** | Formative and summative assessment feedback will be made available verbally and/or in written form. Feedback will be provided within the terms of the University’s guidelines which are provided in the Module Guides. |

© 2026 - The University of Bolton